

1. Compiler vs Interpreter in Food Delivery Application

For a food delivery application, a compiler is preferred during deployment because compiled programs execute faster. A compiler converts the entire source code into machine code before execution, which improves performance and reduces execution time.

Advantages of Compiler:

- Faster execution speed
- Better error detection before program runs
- More secure deployment
- No need for source code during execution

Interpreter Usage:

An interpreter executes code line by line. It is useful during development and testing because errors can be identified quickly, but execution is slower compared to compiled programs.

2. Programming Constructs for Employee Salary Calculation

To calculate employee salary with conditions like bonus, tax, and overtime, multiple programming constructs should be combined.

Constructs Used:

1. **Variables**
Store salary details such as basic salary, bonus, overtime, and tax.
2. **Conditional Statements (if-else)**
Apply conditions for bonus, tax deduction, or overtime payment.
3. **Arithmetic Operators**
Perform salary calculations.
4. **Functions**
Divide the program into smaller reusable parts like tax calculation or bonus calculation.

This combination makes the program efficient, organized, and easier to maintain.

3. Why Loops are Preferred in Menu-Driven Programs

Loops are preferred in menu-driven programs because the program needs to run repeatedly until the user chooses to exit.

Benefits of Loops:

- Avoid writing repetitive code
- Allow continuous user interaction
- Make programs efficient and manageable

Best Loop Choice:

The `do-while` loop is commonly used because:

- The menu is displayed at least once
- The loop continues until the exit condition becomes false

This is ideal for console applications with repeated operations.

4. Incorrect Average Marks Due to Data Type

The probable cause is the use of the `int` data type for average calculation.

Problem:

The `int` type stores only whole numbers and removes decimal values during division.

Example:

If the average is 86.67, using `int` will store only 86.

Solution:

Use `float` or `double` data types because they can store decimal values accurately.

Benefit:

- Correct average calculation
- Better precision in mathematical operations

5. Program Compiles but Crashes at Runtime

This is called a **Runtime Error**.

Meaning:

The program has no syntax errors and compiles successfully, but it fails while executing.

Common Causes:

- Division by zero
- Invalid memory access
- File handling issues
- Null pointer access

Debugging Practices:

1. **Use Debugging Tools**
 - Check program execution step by step using breakpoints.
2. **Print Variable Values**
 - Use print statements to identify incorrect values or program flow issues.

Additional practices include validating inputs and handling exceptions properly.

6. Why Structure is Better Than Multiple Arrays

A structure is preferred because it stores related data together in a single unit.

Example:

For student records:

- Name
- Marks
- Grade

All these values can be grouped into one structure.

Advantages of Structure:

- Better organization of data
- Easier to manage records
- Improves code readability
- Reduces confusion compared to using separate arrays

Structures are especially useful when handling multiple records with related information.